📘 AI/ML Internship – Day 44 Notes & Tasks

🟢 Module 6: Introduction to YOLO & Computer Vision

📌 Part 1: What is Computer Vision?

🔹 Definition

Computer Vision is a field of Artificial Intelligence that enables computers to understand and analyze images and videos.

Just like humans use their eyes to see objects, computers use Computer Vision to identify and understand visual information.

🔹 Real-Life Examples

Face Unlock in Mobile Phones

👉 Detects your face and unlocks the phone.

Self-Driving Cars

👉 Detects:

- Cars
- Pedestrians
- Traffic Signals

CCTV Surveillance

👉 Detects people and suspicious activities.

Medical Imaging

👉 Detects diseases from X-rays and MRI scans.

📌 Part 2: What is Object Detection?

Object Detection means:

1. Identifying an object.
2. Finding its location in an image.

Example

Suppose an image contains:

- Dog
- Cat
- Car

Object Detection will identify all three objects and draw boxes around them.

```
+---------+
|   Dog   |
+---------+

+---------+
|   Cat   |
+---------+

+---------+
|   Car   |
+---------+
```

📌 Part 3: What is YOLO?

🔹 Full Form

YOLO = You Only Look Once

🔹 Definition

YOLO is a popular Object Detection algorithm used in Computer Vision.

It can:

✅ Detect objects

✅ Classify objects

✅ Locate objects

in a single pass.

📌 Part 4: Why is YOLO Popular?

Before YOLO:

- Object detection was slow.
- Multiple scans were needed.

YOLO:

- Looks at image only once.
- Detects multiple objects simultaneously.
- Very fast and accurate.

📌 Part 5: Theory Answers (IMPORTANT)

1. What is Computer Vision?

Computer Vision is a branch of AI that enables computers to understand images and videos.

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2. What is Object Detection?

Object Detection identifies objects and their locations in an image.

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3. What is YOLO?

YOLO is a real-time object detection algorithm.

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4. What is the full form of YOLO?

You Only Look Once.

---

5. Why is YOLO fast?

Because it processes the image in a single pass.

📌 Part 6: Image Classification vs Object Detection

Image Classification

Only identifies what is present.

Example:

Image → Dog

Object Detection

Identifies object and location.

Example:

Dog → Box Position

Feature	Classification	Object Detection
Detect Object	Yes	Yes
Locate Object	No	Yes
Bounding Box	No	Yes

📌 Part 7: Bounding Box

What is a Bounding Box?

A rectangle drawn around an object.

Example:

```
-----------------
|     DOG       |
-----------------
```

Bounding boxes help identify object locations.

📌 Part 8: YOLO Workflow

Image
↓
YOLO Model
↓
Detect Objects
↓
Draw Bounding Boxes

↓
Display Results

📌 Part 9: Real-World Applications of YOLO

🚗 Self-Driving Cars

Detect:

- Cars
- People
- Traffic Lights

🏥 Healthcare

Detect:

- Tumors
- Diseases

🛒 Retail Stores

Detect:

- Products
- Customers

🎥 Security Systems

Detect:

- Intruders
- Suspicious Activities

🏭 Manufacturing

Detect:

- Defective Products

📌 Part 10: YOLO Versions

Popular versions:

YOLOv3
YOLOv4
YOLOv5
YOLOv7
YOLOv8
YOLOv11

Currently, YOLOv8 and newer versions are widely used in projects.

📌 Part 11: Install YOLO

Install Ultralytics YOLO:

```
pip install ultralytics
```

Check installation:

```
yolo checks
```

📌 Part 12: Import YOLO

```
from ultralytics import YOLO
```

📌 Part 13: First YOLO Program

```
from ultralytics import YOLO

model = YOLO("yolov8n.pt")
```

Explanation

yolov8n.pt

Pre-trained YOLO model.

Already trained on thousands of images.

📌 Part 14: Run Object Detection

```
from ultralytics import YOLO

model = YOLO("yolov8n.pt")

results = model("image.jpg")

results[0].show()
```

Output

YOLO will detect:

Person
Dog
Car
Bus
Bicycle

and draw bounding boxes.

📌 Part 15: Important Terms

Object

The item being detected.

Example:

Dog
Cat
Car
Person

Bounding Box

Rectangle around object.

Confidence Score

Confidence of prediction.

Example:

Dog = 95%

Meaning:

Model is 95% confident.

📌 Part 16: Advantages of YOLO

✅ Very Fast

✅ Real-Time Detection

✅ High Accuracy

✅ Detect Multiple Objects

✅ Easy Deployment

📌 Part 17: Disadvantages of YOLO

❌ May miss very small objects

❌ Requires powerful hardware for large projects

❌ Custom training needs large datasets

📌 Part 18: Interview Questions

What is YOLO?

YOLO is a real-time object detection algorithm.

What does YOLO stand for?

You Only Look Once.

Difference between Classification and Detection?

Classification identifies object type.

Detection identifies object type and location.

What is a Bounding Box?

A rectangle around a detected object.

Why is YOLO fast?

It processes the image only once.

🎯 **Tasks for Day 44**

🧠 **Theory**

1. What is Computer Vision?
2. What is Object Detection?
3. What is YOLO?
4. What is a Bounding Box?
5. Advantages of YOLO?

💻 **Practical**

Task 1

Install YOLO:

pip install ultralytics

Task 2

Import YOLO.

Task 3

Load YOLOv8 model.

Task 4

Research YOLOv3, YOLOv5 and YOLOv8.

Task 5

List 10 real-world applications of YOLO.